Game 2 Find the Treasure

Before you start

These are teacher-designed Scratch 3 instructions. Work through the core game first, then test it with a partner. Save your project in the class folder.

Set up a simple route

Choose a player sprite and a treasure sprite. Draw a clear route on the backdrop with a start and finish. Keep the first route wide and short.

Select the player. Put it at the start, then add a reset script using its starting x and y values. For example:

```
when green flag clicked
go to x: (-180) y: (-100)

when [right arrow] key pressed
change x by (10)

when [left arrow] key pressed
change x by (-10)

when [up arrow] key pressed
change y by (10)

when [down arrow] key pressed
change y by (-10)
```

Add a finish response

Place the treasure at the finish. In the player's green-flag script, after the reset, add:

```
forever
  if <touching [Treasure] ?> then
    say [You found it!] for (2) seconds
    stop [this script]
  end
end
```

Select the actual name of your treasure sprite in the touching block.

Test and improve

Use the arrow keys to reach the treasure. Press the green flag to restart from the same place. Ask a partner to follow the route.

The first version uses an on-screen route: it does not stop a player crossing a wall. Follow the route honestly. Collision walls are an optional teacher-led extension.

If movement is too large, reduce 10 to 5. Explain which blocks move sideways and which move up or down.

Sources and related skills

Related book pages are included after this guide. They teach supporting skills; this game is a separate classroom project.

Official Scratch project ideas: https://scratch.mit.edu/ideas

4.2 Command your sprite

In this lesson

You will learn:

➔ how to add commands to your program.

Load program file

Last lesson you saved your work as a file. Now you will load the file. Load means get a file from storage.



Open the file menu. Click on 'load from your computer'. Choose the file you made last time.

Coloured dots and blocks

You will make a Scratch program. A program is made of **commands**. Commands tell the computer what to do.

Scratch commands are blocks. You fit the blocks together. That makes the program.

Different types of blocks are different colours.

- 'Motion' blocks are blue.
- 'Sound' blocks are pink.

Look for the coloured dots on the left of the screen. Choose a colour to help you pick the blocks you need.

Actions

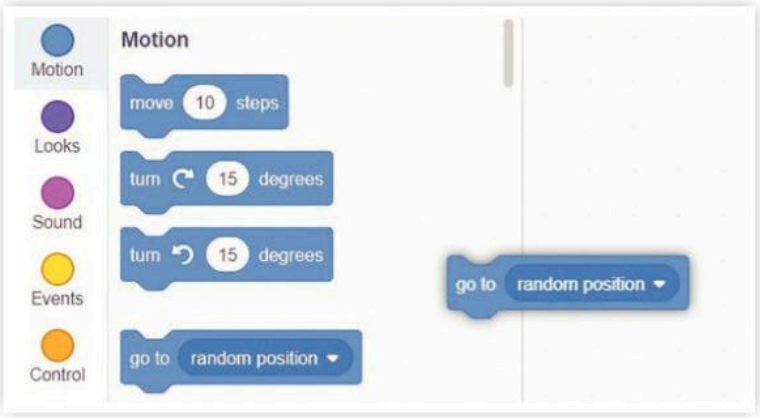
Choose a block. It will make your sprite move.

Click on the blue dot. You will see 'Motion' blocks.

Find the block that says 'go to random position'.

Drag this block to the middle of the screen.

Click on the block. The sprite will move.



Sounds

Choose a block. It will play a sound. Click on the pink dot. You will see 'Sound' blocks. Choose the 'play sound' block.

Now you have two blocks. Fit the blocks together.

Click on the two blocks. What happens?





Activity

Make the two-block program shown on this page.



Extra challenge

Add any other block to the program. See what happens when you run the program.

Think again

Say what blocks you used. Say what each block did.

4.3 A start event

In this lesson

You will learn:

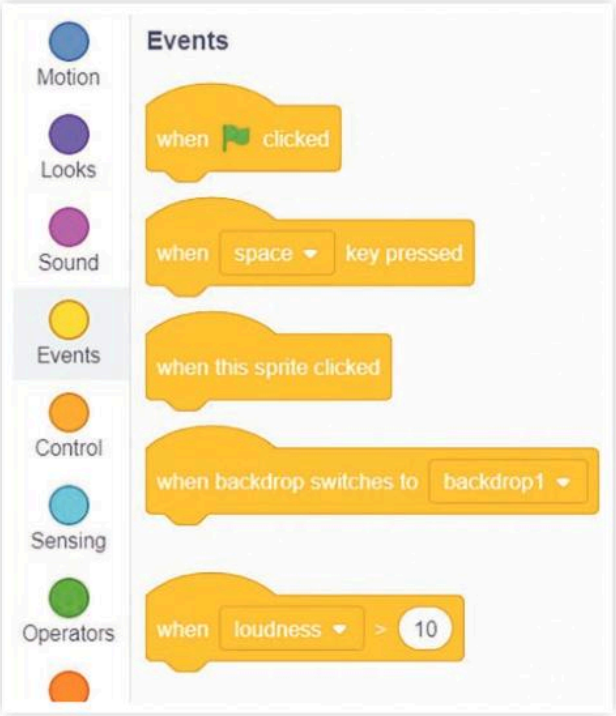
- to choose an event that starts the program.

Events

An **event** means anything that happens. In Scratch 'Events' are things the **user** does. The user is the person who uses the program. Now you will choose the 'Event' that lets the user start your program.

Click on the yellow dot that says 'Events'.

What 'Event' blocks can you see? What does each one say?

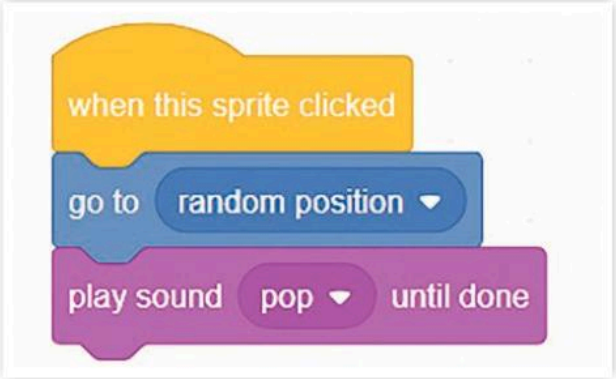


Choose an event

To choose an 'Event' drag the block to the program area.

Find the 'Event' block that says 'when this sprite clicked'.

Fit the program blocks onto the 'Event' block. Now the event will start the program.



Change colour

Now you will make the sprite change colour. Click on the purple dot that says 'Looks'. These command blocks change the look of the sprite.

Find the block that says 'change color effect by…'.

Drag this block to the script area and add it to the program.





Activity

Change your program.

- Add the 'Event' block which says 'when this sprite clicked'.
- Add the 'Looks' block which changes the colour of the sprite.

Click the sprite to make it move.

Save the file.



Extra challenge

Choose a different start event. Change the program to the new event.

Think again

Draw the 'Event' block you used. Say what it does.

If you used more than one, draw them all.

4.5 Plan and make

In this lesson

You will learn:

➔ how to make a new program to match a plan.

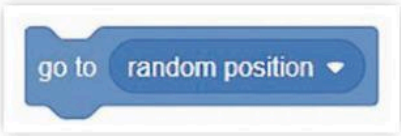
Plan

Here is the plan.

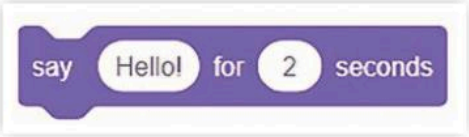
- The sprite will jump to a random place.
- The sprite will say "Hello!"
- The sprite will keep moving towards the mouse pointer.

First commands

The sprite will start in a **random** place. Random means you do not know where it will go.



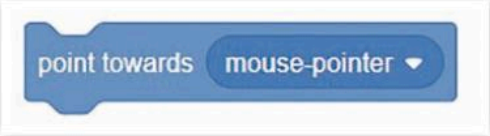
The sprite will say "Hello!"



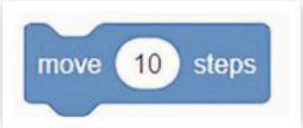
Find these blocks and fit them together.

Repeated commands

Here is the block that makes the sprite point towards the mouse pointer.



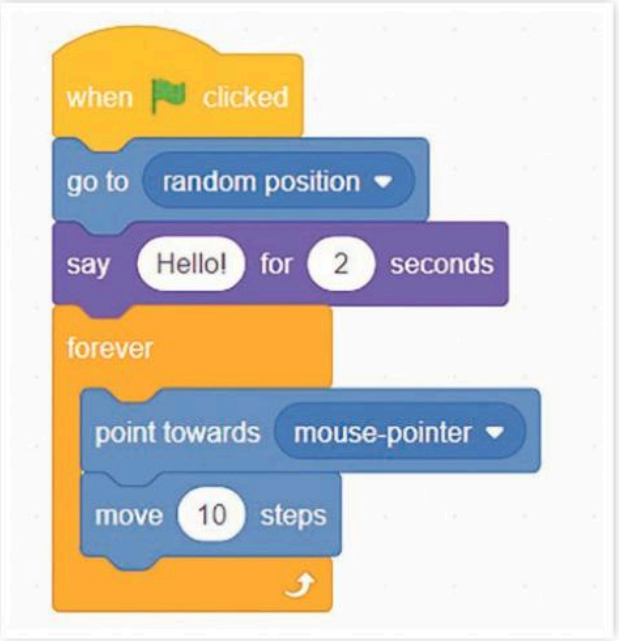
Here is the block to move 10 steps.



These two blocks will go inside a forever loop.

Completed program

Here is the program with all the pieces in place. It has a start 'Event' – the green flag.





Activity

Start a new program. Choose a sprite that looks like a fish. Choose a background that looks like underwater.

Make the program shown on this page. The program will make the fish swim about underwater.

Think again

Think of one change you could make to the program you made. What would happen if you made this change?



Extra challenge

Add a jellyfish sprite to the underwater scene. Make a program for this sprite.

- Use the same blocks as before.
- Take out the block that says 'follow the mouse pointer'.
- Add a block that says 'if on edge bounce'.
- Make the number of steps 2 instead of 10.

